

SC3102: Signals, systems and transforms

Tutorial 2: Signals 2 (Representation of signals, Singularity signals)

1. Sketch the single-sided and double-sided amplitude and phase spectra for the following signals.

(a) $x_1(t) = 10\cos\left(4\pi t + \frac{\pi}{8}\right) + 6\sin\left(8\pi t + \frac{3\pi}{4}\right)$

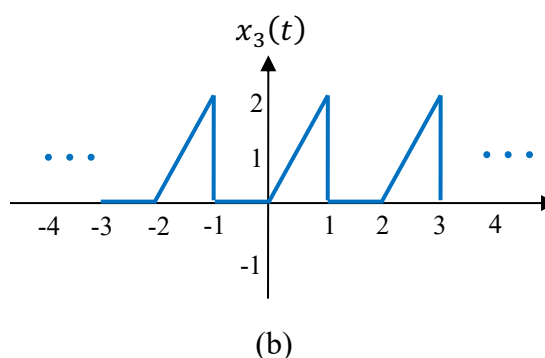
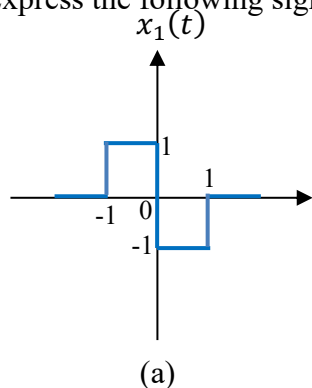
(b) $x_2(t) = 8\sin(3\pi t)\cos(5\pi t)$

2. Sketch the following signals. (Note: The signals $u(t)$, $r(t)$ and $\Pi(t)$ represent unit step, unit ramp and rectangular signals, respectively.)

(a) $x_1(t) = r(t) - r(t - 1)$

(b) $x_2(t) = \sum_{n=-\infty}^{+\infty} g(t - 3n)$, where $g(t) = u(t)u(2 - t)$

3. Express the following signals in terms of singularity functions:



4. Evaluate the following integrals:

(a) $\int_{-\infty}^7 t^2 \delta(t - 3) dt$

0

(b) $\int_{-\infty}^{\infty} \cos(\pi t) \delta(2t - 2) dt$

5. A periodic signal $y(t) = K\cos(6000\pi t)\cos(2000\pi t)$ is transmitted through a digital communication system.

- (a) What is the maximum possible time interval between successive samples for the accurate reconstruction of the signal at the receiver?
- (b) Using the sampling rate obtained above, determine the number of samples needed to store one minute of the signal.
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Answers:

1. (a) Plot the spectra from the following expression

$$x_1(t) = 10\cos\left(4\pi t + \frac{\pi}{8}\right) + 6\cos\left(8\pi t + \frac{\pi}{4}\right)$$

$$= 5e^{j\left(4\pi t + \frac{\pi}{8}\right)} + 5e^{-j\left(4\pi t + \frac{\pi}{8}\right)} + 3e^{j\left(8\pi t + \frac{\pi}{4}\right)} + 3e^{-j\left(8\pi t + \frac{\pi}{4}\right)}$$
 (b) Plot spectra from the following expression

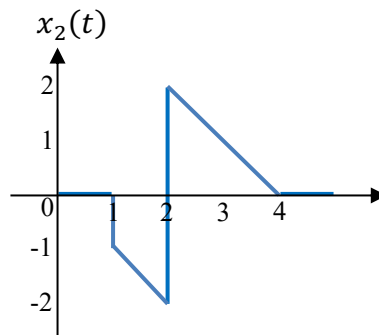
$$x_2(t) = 4\cos\left(8\pi t - \frac{\pi}{2}\right) - 4\cos\left(2\pi t - \frac{\pi}{2}\right)$$

$$= 2e^{j\left(8\pi t - \frac{\pi}{2}\right)} + 2e^{-j\left(8\pi t - \frac{\pi}{2}\right)} - 2e^{j\left(2\pi t - \frac{\pi}{2}\right)} - 2e^{-j\left(2\pi t - \frac{\pi}{2}\right)}$$
2. Sketch the corresponding signals
3. (a) $x_1(t) = u(t+1) - 2u(t) + u(t-1)$
 (b) $x_3(t) = \sum_{n=-\infty}^{\infty} g(t-2n)$ where $g(t) = r(2t)u(1-t)$
4. (a) 9 since $t = 3 \in (-\infty, 7]$
 (b) $-\frac{1}{2}$
5. (a) Maximum possible time interval between samples $T_s = \frac{1}{F_s} = \frac{1}{8000}$ sec
 (b) Number of samples for one minute of signal = 480000 samples

Exercises:

1. Sketch the single-sided and double-sided amplitude and phase spectra for the signal

$$x_3(t) = 4\sin(8\pi t) + 2\sin\left(12\pi t - \frac{\pi}{4}\right) - 10\cos\left(16\pi t + \frac{5\pi}{4}\right).$$
2. Sketch the signal $x_3(t) = \Pi\left(\frac{t-3}{6}\right) + \Pi\left(\frac{t-4}{2}\right)$ and express the derivative of the signal given in in terms of unit impulses.
3. Express the following signal in terms of singularity functions:



4. Evaluate the integral $\int_0^{\infty} \sqrt{t+3} \delta(t+2) dt$

Answers (Exercise):

1. Plot the spectra from the following expression

$$\begin{aligned}x_3(t) &= 4\cos\left(8\pi t - \frac{\pi}{2}\right) + 2\cos\left(12\pi t - \frac{3\pi}{4}\right) - 10\cos\left(16\pi t + \frac{5\pi}{4}\right) \\&= 2e^{j(8\pi t - \frac{\pi}{2})} + 2e^{-j(8\pi t - \frac{\pi}{2})} + e^{j(12\pi t - \frac{3\pi}{4})} + e^{-j(12\pi t - \frac{3\pi}{4})} \\&\quad - 5e^{j(16\pi t + \frac{5\pi}{4})} - 5e^{-j(16\pi t + \frac{5\pi}{4})}\end{aligned}$$

2. Sketch the corresponding signal. The derivative of $x_3(t)$ is

$$\frac{d}{dt}x_3(t) = \delta(t) + \delta(t - 3) - \delta(t - 5) - \delta(t - 6)$$

3. $x_2(t) = -u(t - 1) + 4u(t - 2) - r(t - 1) + r(t - 4)$

4. 0 since $t = -2 \notin [0, +\infty)$
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